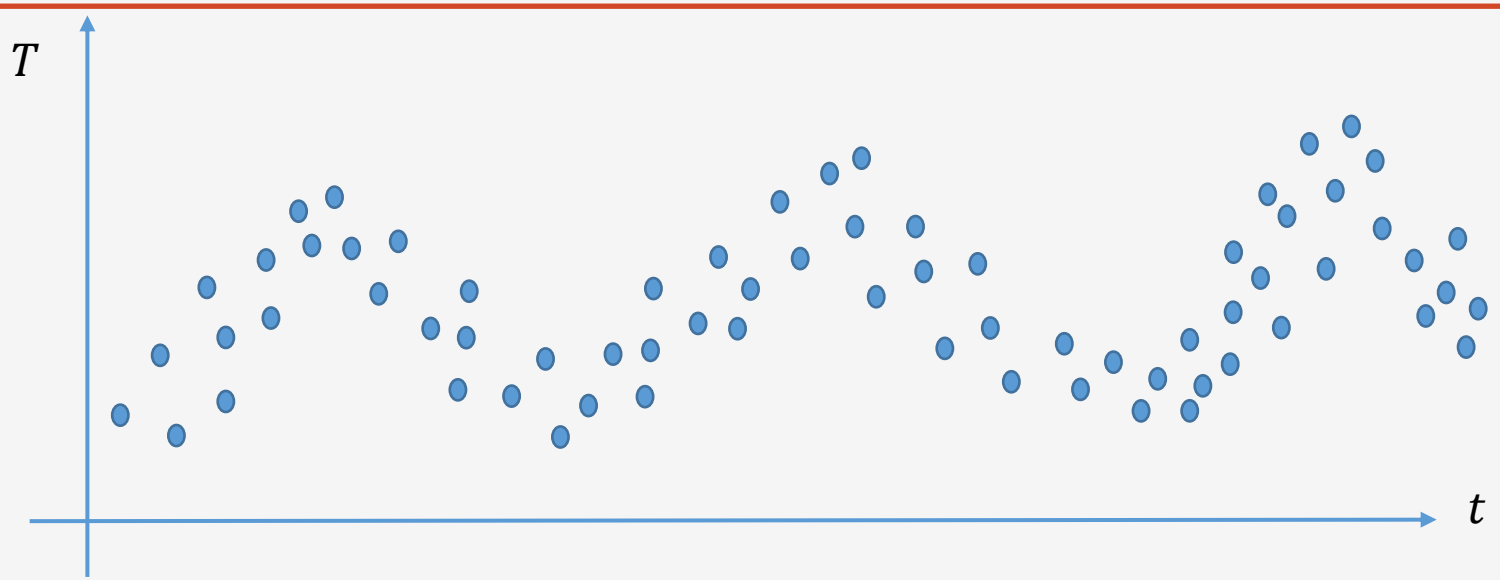
Gauss-Newton method (Lec9.pdf)

Ioannis Akrotirianakis

Todays topics & Logistics

- Nonlinear Least Squares
- Gauss-Newton method for solving NLS problems

Let's start with an example



- We have temperature measurements (T_i) over time points $t_i \in [t_{min}, t_{max}]$
- The dataset looks like this $D = \{(T_i, t_i): i = 1,2, \dots, m\}$; (total of m measurements)
- **Questions: (1)** Can we use Linear Regression? **(2)** How well it will fit the data??

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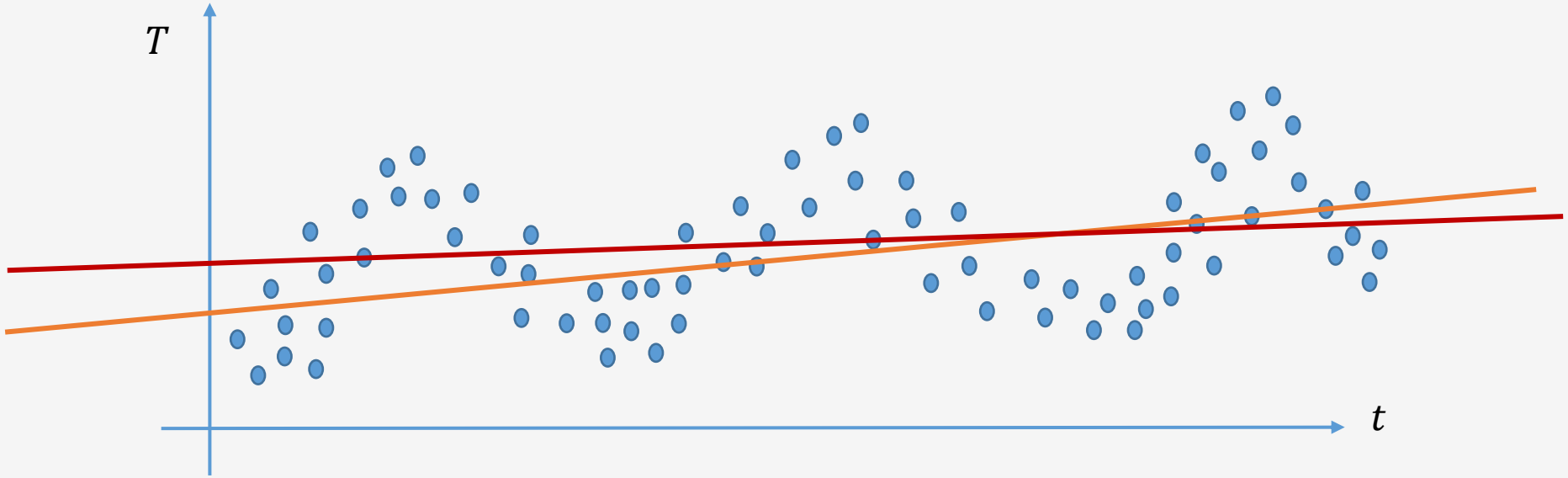

- We have temperature measurements (T_i) over a period of time $t \in [t_{min}, t_{max}]$
- The dataset looks like this $D = \{(T_i, t_i): i = 1,2, \dots, m\}$; total of m measurements
- **Questions: (1)** Can we use Linear Regression? **(2)** How well it will fit the data??

YES NOT GOOD FIT

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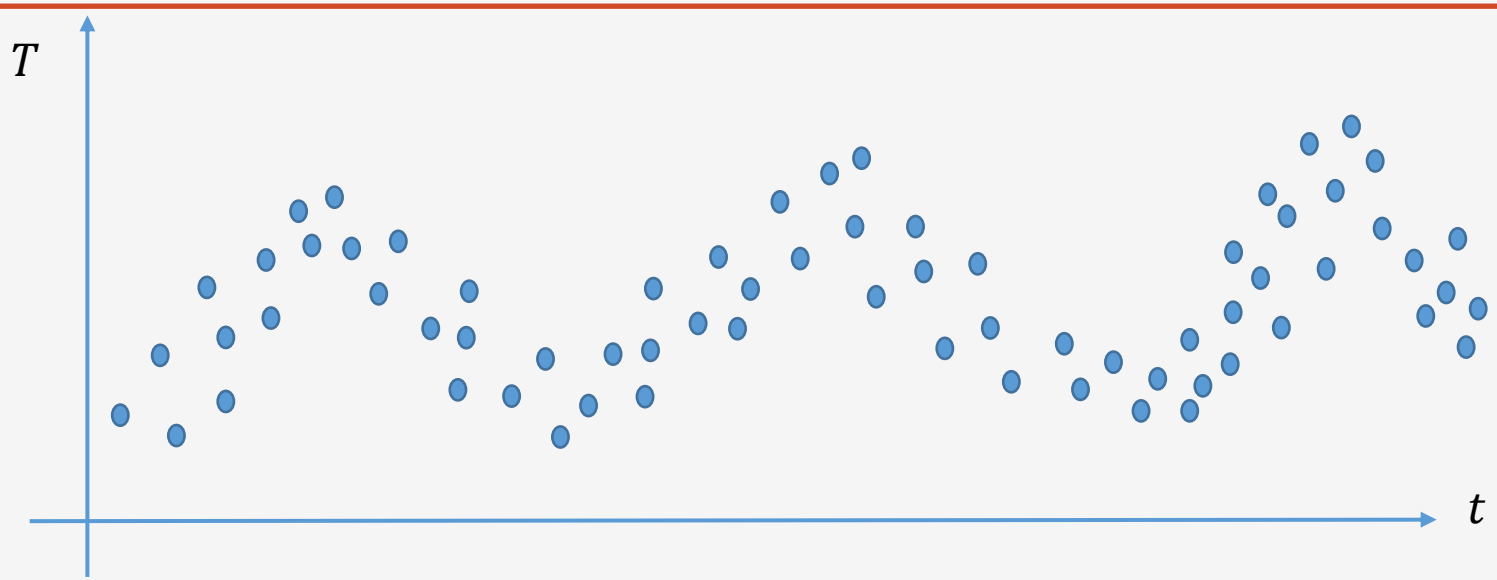

- What observations can we make by looking at the data ??

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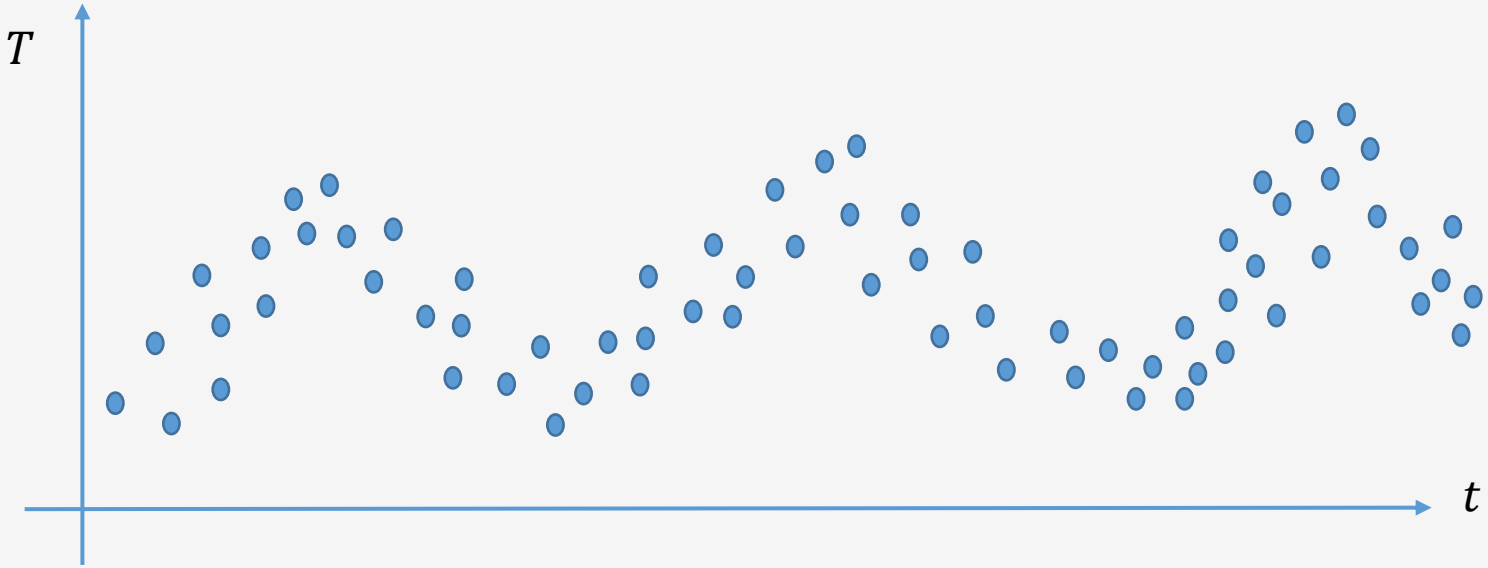

- What observations can we make by looking at the data ??
  - There is seasonality (i.e., periodicity)
  - There is a slight increasing trend

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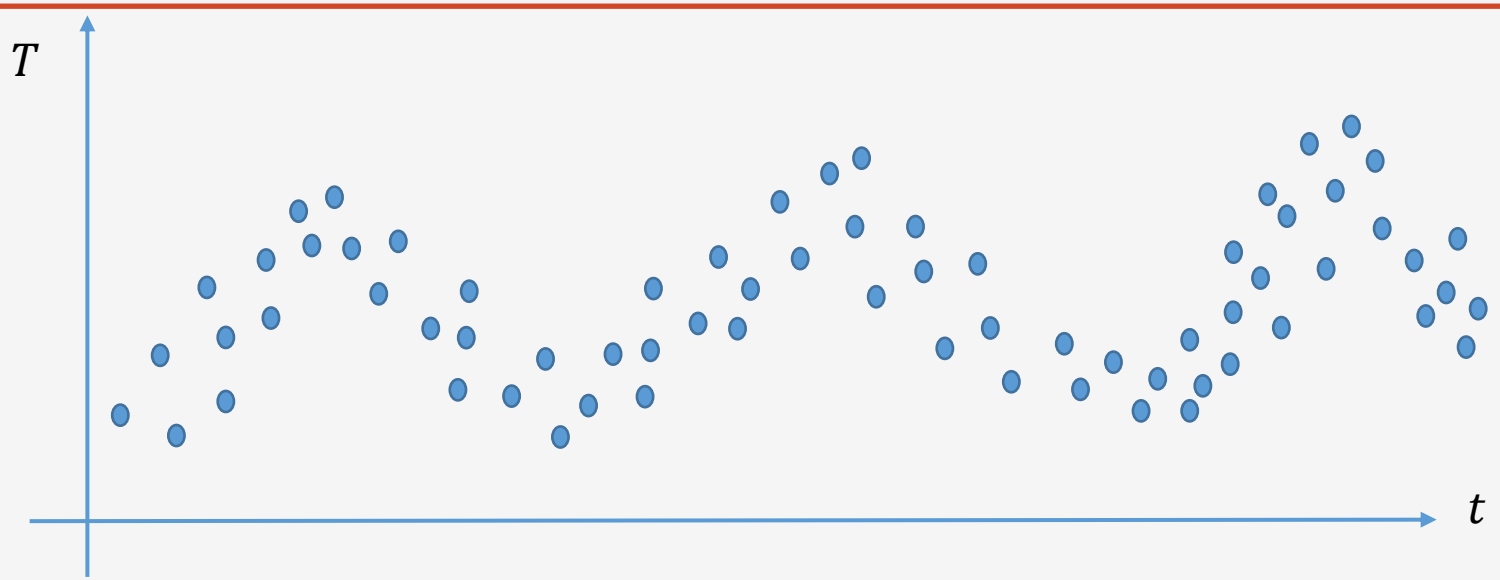

- What observations can we make by looking at the data ??
  - There is seasonality (i.e., periodicity) → what functions come to mind??
  - There is a slight increasing trend → what functions come to mind??

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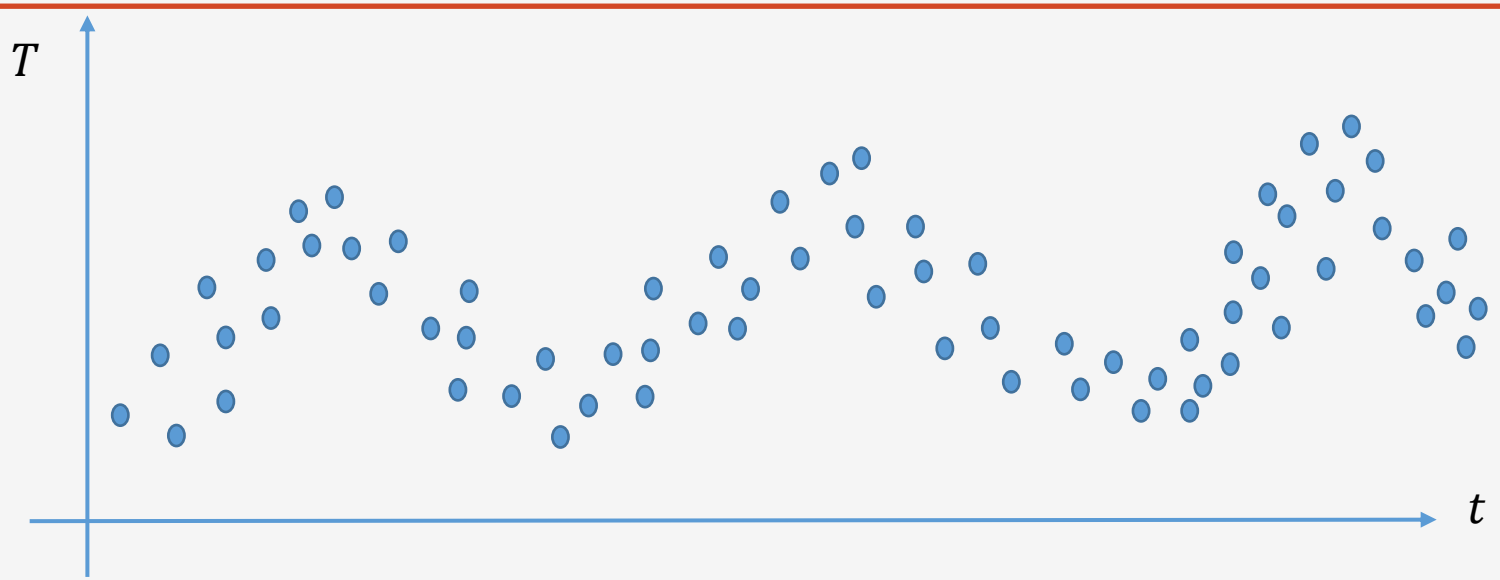

- What observations can we make by looking at the data ??
  - There is seasonality (i.e., periodicity) → trigonometric (sine, cosine, …)
  - There is a slight increasing trend → affine functions with increasing slope

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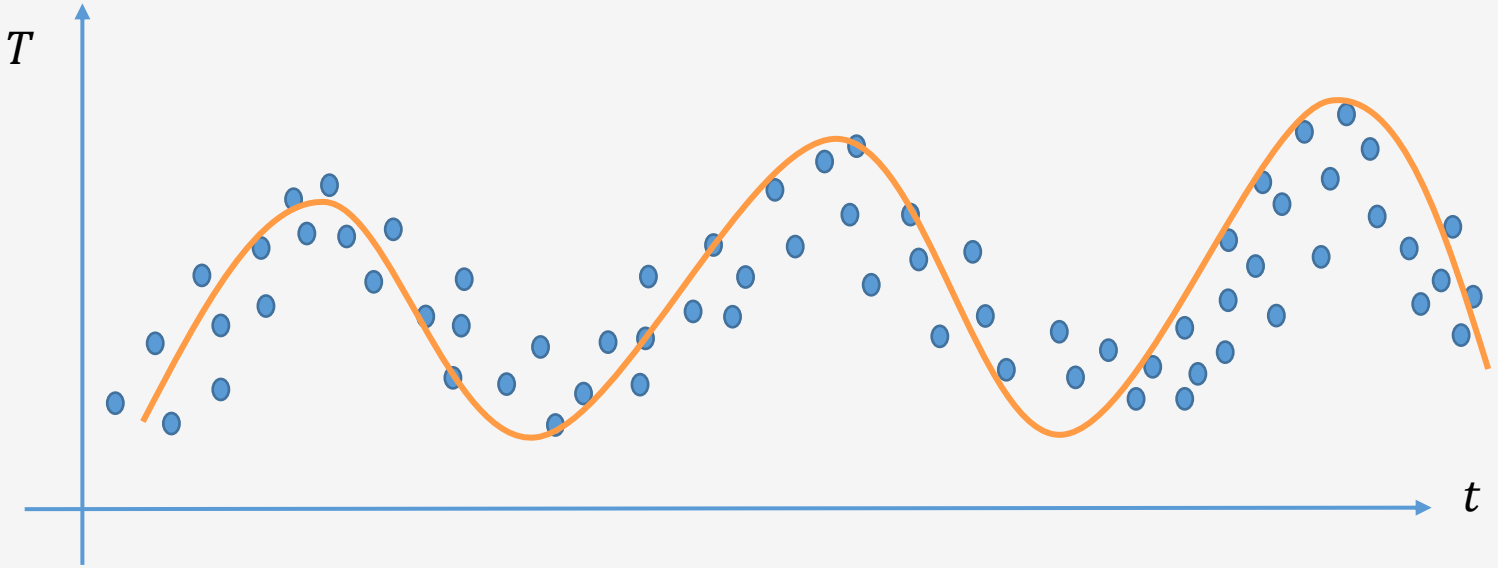

- A possible model can have the form: $\boldsymbol{\mathcal{L}(a, \omega, \phi, b; t) = a \sin(\omega t + \phi) + bt}$
- Seasonality/periodicity term: $a \sin(\omega t + \phi)$
- Slight increasing trend: bt

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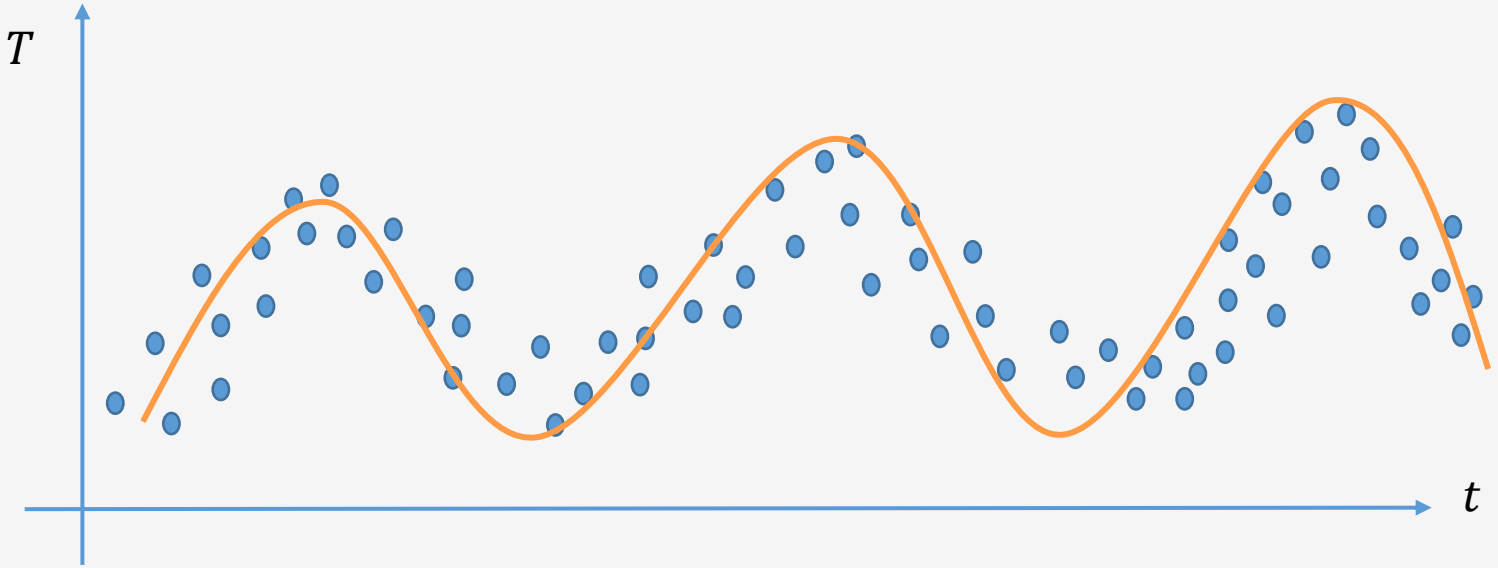

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- What are the data and the decision variables in the above model **??**

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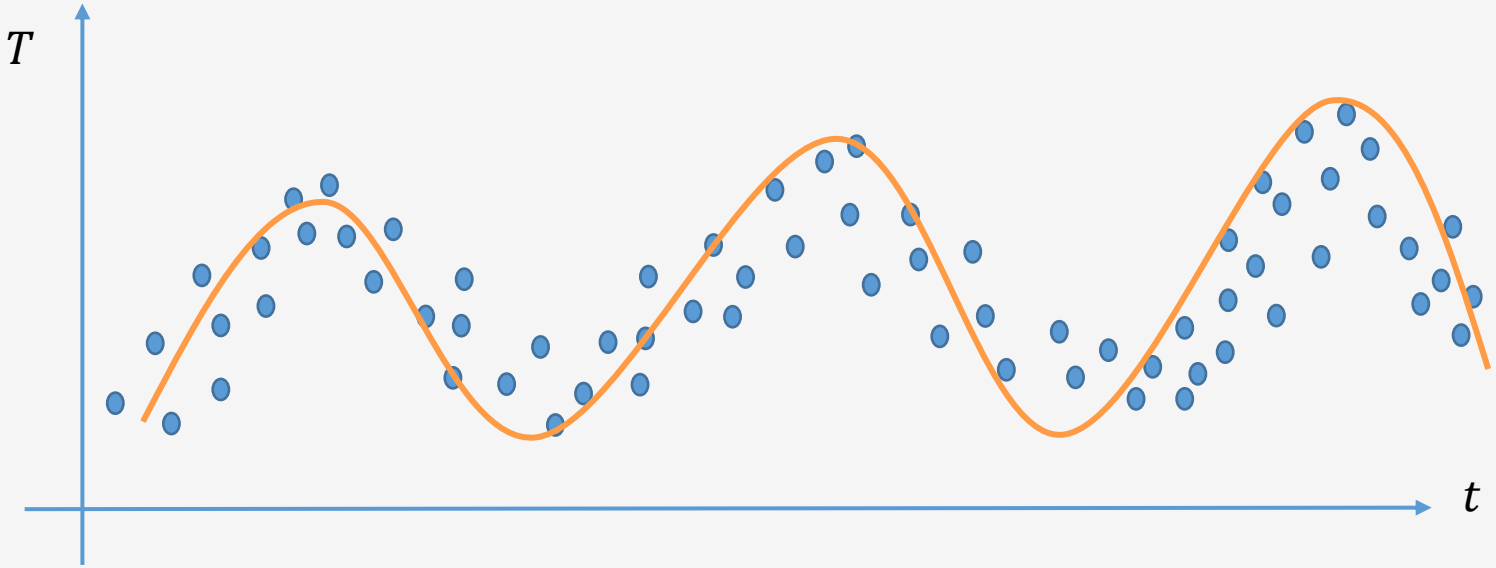

- A possible model can have the form: $\mathcal{L}(a, \omega, \phi, b; t) = a \sin(\omega t + \phi) + bt$
- What are the data and the decision variables in the above model **??**
- Data: t (the times we made the measurements); Variables: a, ω, ϕ, b

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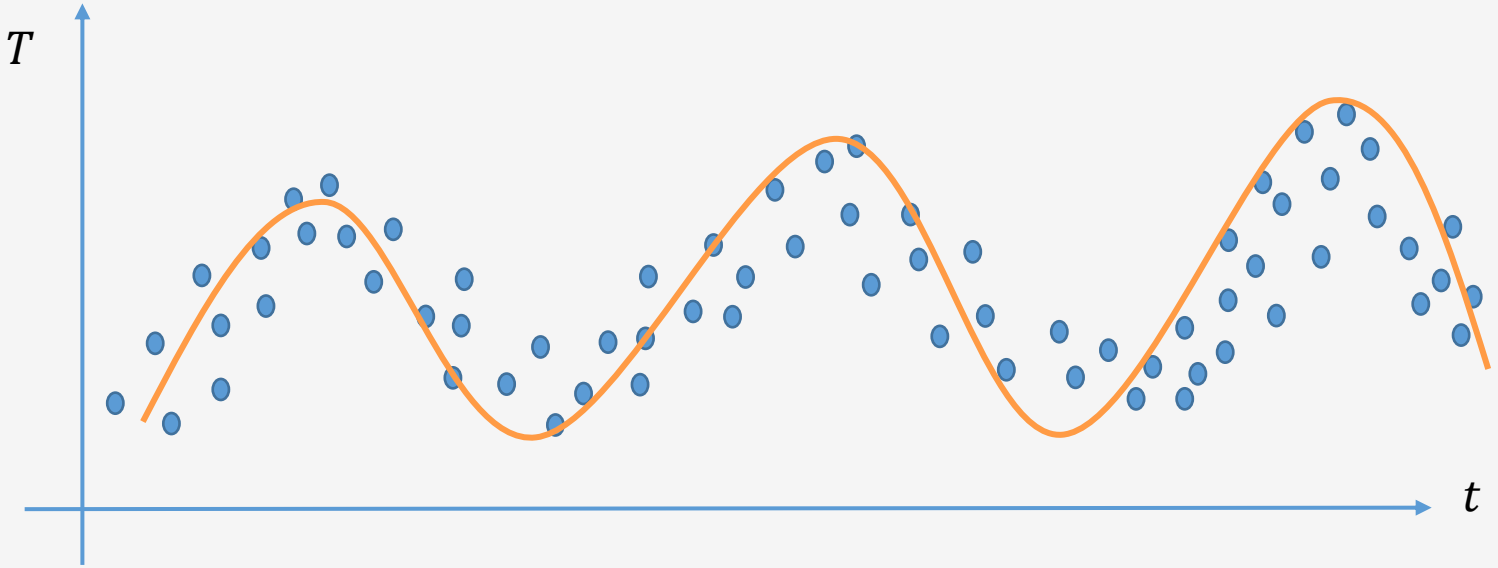

- A possible model can have the form: $\mathcal{L}(a, \omega, \phi, b; t) = a \sin(\omega t + \phi) + bt$

- What is the main purpose of the model?? Why do we want to compute (learn) it??

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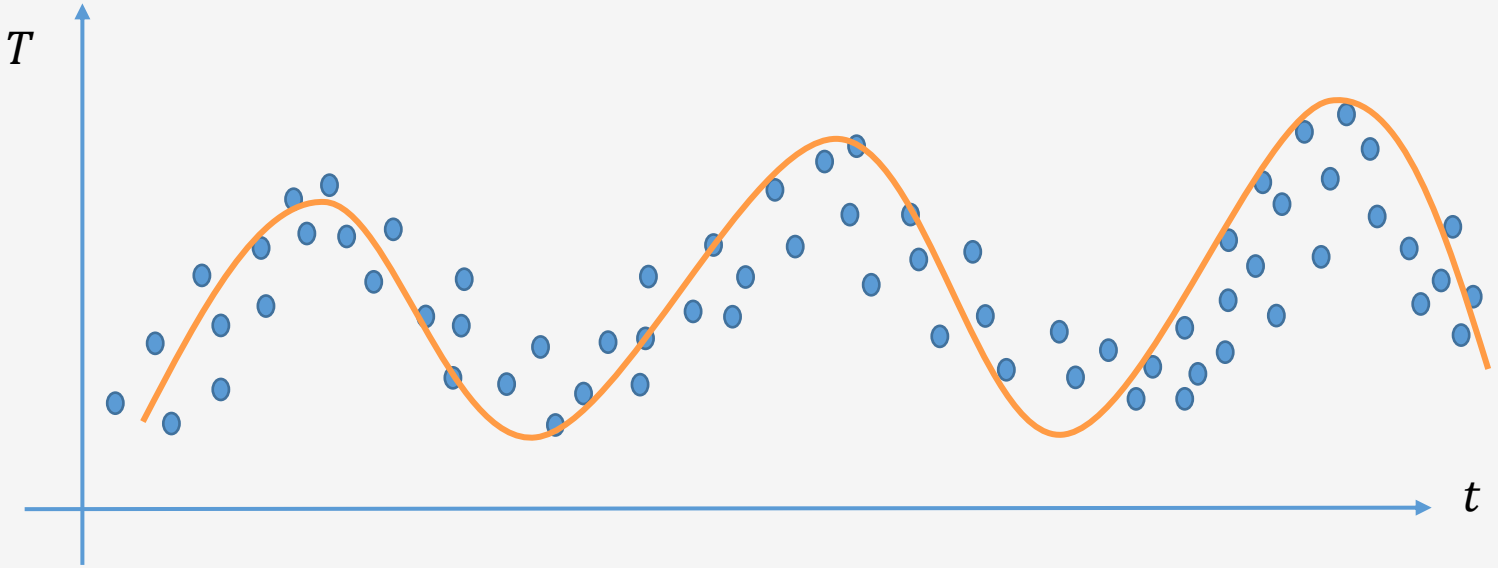

- A possible model can have the form: $\mathcal{L}(a, \omega, \phi, b; t) = a \sin(\omega t + \phi) + bt$
- What is the main purpose of the model?? Why do we want to compute (learn) it??
- We want to use it to estimate Temperatures in Future time points

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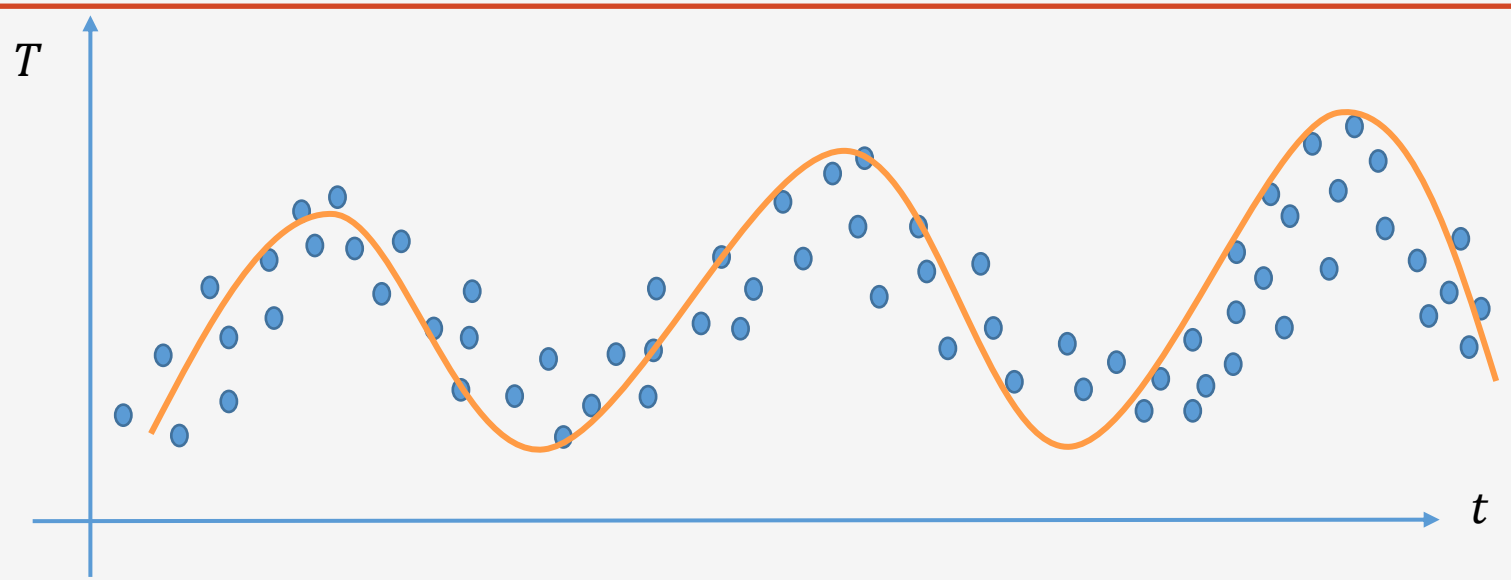

- Our **goal/objective** is to compute (learn) the best (optimal) values for a, ω, ϕ, b such that the **deviations**

Actual Temperature at time t_i

$$g_i(a, \omega, \phi, b) = \left(T_i - (a \sin(\omega t_i + \phi) + b t_i)\right)^2 = \left(T_i - \underbrace{\mathcal{L}(a, \omega, \phi, b; t_i)}_{\text{Model predicting Temperature at time } t_i}\right)^2$$

- are **minimized (abs)**, for all measurements $(\mathrm{T_i}, t_i)$ $i = 1, \ldots, m$

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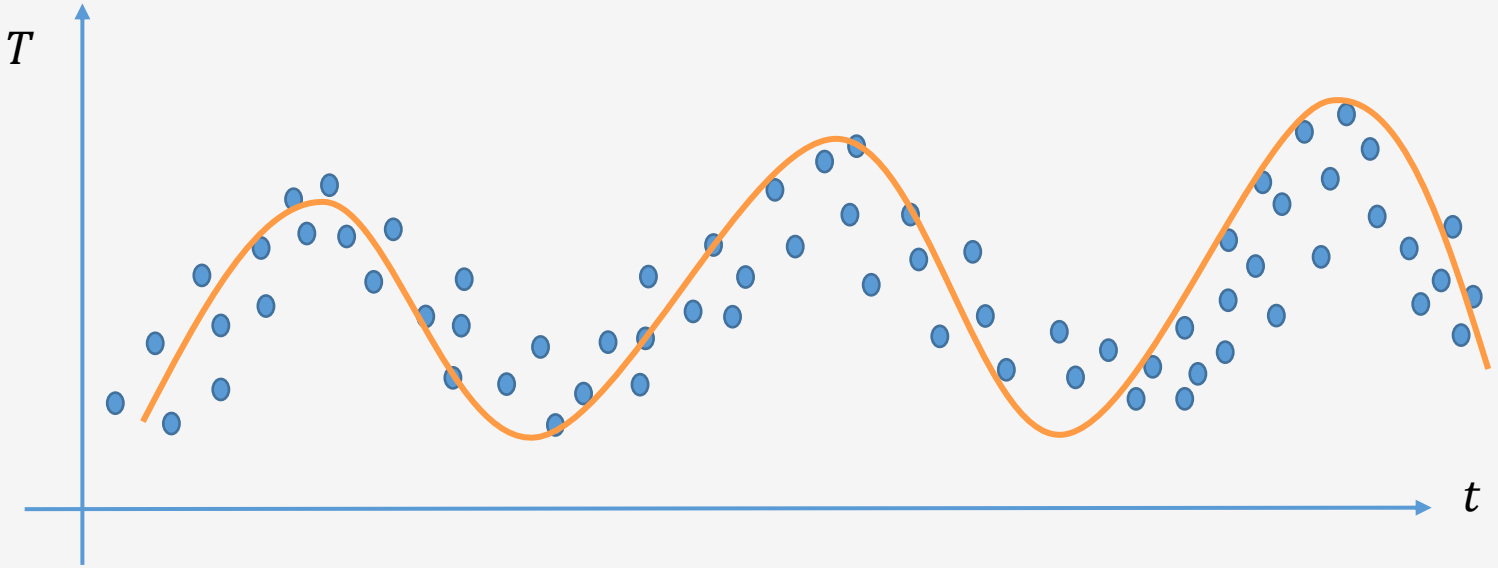
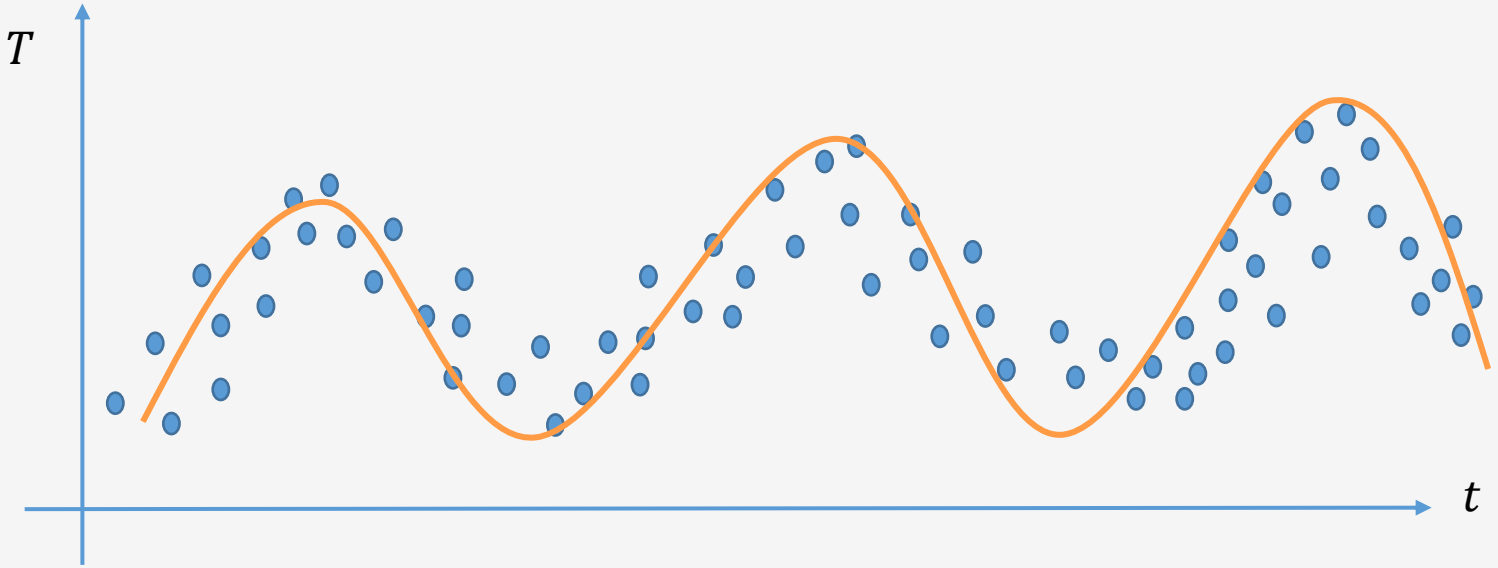


- Hence, we consider the unconstrained optimization problem

$$\min \mathrm{f}(\mathrm{a}, \omega, \phi, b) = \sum_{i=1}^{m} (T_i - a\sin(\omega t_i + \phi) - bt_i)^2$$

Let's start with an example



- Hence, we consider the unconstrained optimization problem

$$\min \mathrm{f}(\mathrm{a}, \omega, \phi, b) = \sum_{i=1}^{m} (T_i - a\sin(\omega t_i + \phi) - b t_i)^2 = \sum_{i=1}^{m} \left(g_i(a, \omega, \phi, b)\right)^2$$

Nonlinear Least Squares

- The general formulation of the Nonlinear Least Squares is seeking to minimize the squares of m (nonlinear) functions $g_i: R^n \to R, i = 1,2, \dots, m$

$$\min_{x \in R^n} f(x) = \frac{1}{2} \sum_{i=1}^{m} \left(g_i(x)\right)^2$$

- **Observations**:
  - We have n decision variables ($x \in R^n$)
  - We have m (nonlinear) functions
  - Usually, we have $m \gg n$ (i.e., many more observations than variables)

Nonlinear Least Squares

- Define the **vector function** $g: R^n \to R^m$ with $g(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \\ \dots \\ g_m(x) \end{pmatrix}$

- Then the Nonlinear Least Squares problem can be written as

$$\min_{x \in R^n} f(x) = \frac{1}{2} \sum_{i=1}^{m} \left(g_i(x)\right)^2 = \frac{1}{2} \left(g(x)\right)^T g(x) = \frac{1}{2} \left|\left|g(x)\right|\right|^2$$

Nonlinear Least Squares

- How can we compute the gradient of the vector function $g: R^n \rightarrow R^m$??
- Any intuition how it would look like ??
  - Can it be a vector ??
  - Can it be a matrix ??

The Jacobian matrix

The gradient of the vector function g is called the **Jacobian matrix** of g

$$\nabla g(x)^T = J(x) = \begin{pmatrix} \nabla g_1^T(x) \\ \nabla g_2^T(x) \\ \dots \\ \nabla g_m^T(x) \end{pmatrix} \in R^{m \times n}$$

The (i, j) element of the Jacobian matrix is: $\boldsymbol{J_{ij}} = \frac{\partial g_i(x)}{\partial x_j}$, for $i = 1, \dots m$ and $j = 1, \dots, n$

Gauss-Newton method for Nonlinear Least Squares

The **iterative formula** for solving the Nonlinear Least Squares is:

$$x_{k+1} = x_k - \left(J(x_k)^T J(x_k)\right)^{-1} J(x_k)^T g(x_k)$$

Gauss-Newton method uses only **1st order information** (only 1st derivatives)

What is the direction of movement ??

Gauss-Newton method for Nonlinear Least Squares

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Gauss-Newton method uses only **1st order information** (only 1st derivatives)

Direction of movement: $d_k = -\left(J(x_k)^T J(x_k)\right)^{-1} J(x_k)^T g(x_k)$

Gauss-Newton method for Nonlinear Least Squares

Is $d_k = -\left(J(x_k)^T J(x_k)\right)^{-1} J(x_k)^T g(x_k)$ a **descent direction** ??

Gauss-Newton method for Nonlinear Least Squares

- Is $d_k = -\left(J(x_k)^T J(x_k)\right)^{-1} J(x_k)^T g(x_k)$ a descent direction ?? **YES**
  - Because:
    - $J(x_k)^T g(x_k)$ is the **gradient of the objective**: $f(x) = \frac{1}{2}\left|\left|g(x)\right|\right|^2$
    - The matrix $\left(\boldsymbol{J(x_k)^T J(x_k)}\right) \succ \mathbf{0}$, if $J(x_k)$ has <u>full column rank</u>
      - And in this case, $\left(J(x_k)^T J(x_k)\right)$ is also **invertible**

Derivation of the Gauss-Newton formula

$$\min_{x \in R^n} f(x) = \frac{1}{2}\sum_{i=1}^{m}\left(g_i(x)\right)^2 = \frac{1}{2}\left|\left|g(x)\right|\right|^2$$

- How can we solve the **Nonlinear LS** as an optimization problem ??
- Recall that we know how to solve **Linear LS** problems (when the g_i are affine)

Derivation of the Gauss-Newton formula

$$\min_{x \in R^n} f(x) = \frac{1}{2}\sum_{i=1}^{m} \left(g_i(x)\right)^2 = \frac{1}{2}\left|\left|g(x)\right|\right|^2$$

- How can we solve the above optimization problem ??
- We know how to solve Linear Least Squares problems (when the g_i are affine)
- How about we approximate **each** g_i by affine functions → 1st order Taylor

$$\tilde{g}_i(x) \approx g_i(x_k) + \nabla g_i(x)^T (x - x_k), \qquad \forall i = 1,2,\dots,m$$

Derivation of the Gauss-Newton formula

- For the **vector function** $\tilde{g}(x) = \left(\tilde{g}_1(x), \dots, \tilde{g}_m(x)\right)^T$ the 1st order Taylor becomes:

$$\tilde{g}(x) = g(x_k) + J(x_k) \quad (x - x_k)$$

$$\Updownarrow$$

$$\begin{pmatrix} \tilde{g}_1(x) \\ \tilde{g}_2(x) \\ \dots \\ \tilde{g}_m(x) \end{pmatrix} = \begin{pmatrix} g_1(x_k) \\ g_2(x_k) \\ \dots \\ g_m(x_k) \end{pmatrix} + \begin{pmatrix} \nabla g_1(x_k)^T \\ \nabla g_2(x_k)^T \\ \dots \\ \nabla g_m(x_k)^T \end{pmatrix} (x - x_k)$$

Derivation of the Gauss-Newton formula

- Instead of minimizing the original Nonlinear Least Squares problem, we will **minimize its linear approximation**:

$$\min \tilde{f}(x) = \frac{1}{2}\left|\left|\tilde{g}(x)\right|\right|^2 = \frac{1}{2}\left|\left|g(x_k) + J(x_k)^T(x - x_k)\right|\right|^2$$

- This is **Quadratic Optimization problem** ☺

- If we **expand** the objective function, we get:

$$\tilde{f}(x) = \frac{1}{2}\left(\left|\left|g(x_k)\right|\right|^2 + 2(x - x_k)^T \underbrace{J(x_k)g(x_k)}_{\boldsymbol{b}} + (x - x_k)^T \underbrace{J(x_k)^T J(x_k)}_{\boldsymbol{Q}}(x - x_k)\right) = \frac{1}{2}(x^T Q x + b^T x + c)$$

Derivation of the Gauss-Newton formula

- Using **FONC** on the quadratic function $\tilde{f}$, we get the candidate point $\tilde{x}$:

$$\nabla\tilde{f}(\tilde{x}) = 0 \Rightarrow Q(\tilde{x} - x_k) + b = 0 \Rightarrow Q\tilde{x} - Qx_k + b = 0 \Rightarrow \tilde{x} = x_k - Q^{-1}b \Rightarrow$$

$$\Rightarrow \boxed{\tilde{x} = x_k - \left(J(x_k)^T J(x_k)\right)^{-1} J(x_k) g(x_k)}$$

- We take $\boldsymbol{x_{k+1}} = \boldsymbol{\tilde{x}}$ and we have derived the Gauss-Newton iteration. ■